WAYGROUND formerly Quizizz Worksheets

Name

**FTW_Databricks Data Engineer Associate V3.2**

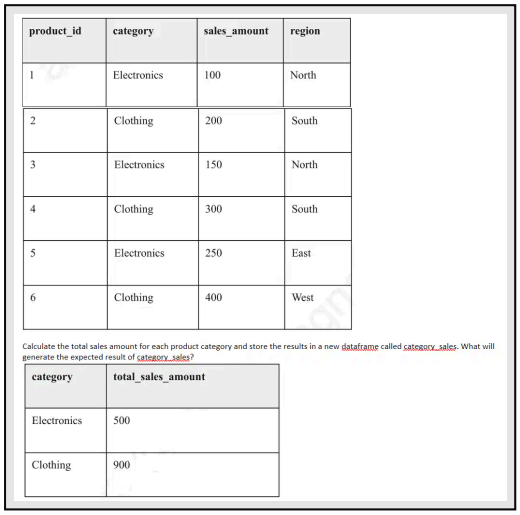
Class

Total questions: 90

Date

Worksheet time: 47mins

1.



| product_id | category | sales_amount | region |
|---|---|---|---|
| 1 | Electronics | 100 | North |
| 2 | Clothing | 200 | South |
| 3 | Electronics | 150 | North |
| 4 | Clothing | 300 | South |
| 5 | Electronics | 250 | East |
| 6 | Clothing | 400 | West |

Calculate the total sales amount for each product category and store the results in a new dataframe called category_sales. What will generate the expected result of category_sales?

| category | total_sales_amount |
|---|---|
| Electronics | 500 |
| Clothing | 900 |

A global retail company sells products across multiple categories (e.g., Electronics, Clothing) and regions (e.g., North, South, East, West).
The sales team has provided the data engineer with a PySpark dataframe named sales_df and the team wants the data engineer to analyze the sales data to help them make strategic decisions.
sales_df (See screenshot)

Calculate the total sales amount for each product category and store the results in a new dataframe called category_sales.
What will generate the expected result of category_sales? (See screenshot)

a)

```
category_sales = sales_df
.groupBy("category")
.agg(sum("sales_amount")
.alias("total_sales_amount"))
```

b)

```
category_sales = sales_df
.sum("sales_amount")
.groupBy("category")
.alias("total_sales_amount"))
```

c)

```
category_sales = sales_df
.agg(sum("sales_amount")
.groupBy("category")
.alias("total_sales_amount"))
```

d)

```
category_sales = sales_df
.groupBy("region")
.agg(sum("sales_amount")
.alias("total_sales_amount"))
```

2.

| product_id | category | sales_amount | region |
|---|---|---|---|
| 1 | Electronics | 100 | North |
| 2 | Clothing | 200 | South |
| 3 | Electronics | 150 | North |
| 4 | Clothing | 300 | South |
| 5 | Electronics | 250 | East |
| 6 | Clothing | 400 | West |

Calculate the total sales amount for each region and store the results in a new dataframe called region_sales. Given the expected result:

| region | total_sales_amount |
|---|---|
| North | 250 |
| South | 500 |
| East | 250 |
| West | 400 |

A company sells products across multiple categories (e.g., Electronics, Clothing) and regions.
The sales team has provided you with a PySpark dataframe named sales_df as below,
and the team wants the data engineer to analyze the sales data to help make strategic decisions.

Calculate the total sales amount for each region and store the results in a new dataframe called region_sales.
Given the expected result: (See image)

Which code will generate the expected result?

a)
```
region_sales = sales_df
.groupBy("category")
.sum("sales_amount")
.alias("total_sales_amount")
```

b)
```
region_sales = sales_df
.groupBy("region")
.agg(sum("sales_amount")
.alias("total_sales_amount"))
```

c)
```
region_sales=sales_df
.sum("sales_amount")
.groupBy("region")
.alias("total_sales_amount")
```

d)
```
region_sales =sales_df
.agg(sum("sales_amount")
.groupBy("region")
.alias("total_sales_amount"))
```

3. A data engineering team is using Kafka to capture event data and then ingest it into Databricks. The team wants to be able to see these historical events. Medallion architecture is already in place. The team wants to be mindful of costs. Where should this historical event data be stored?

a) Gold
b) Silver
c) Bronze
d) Raw layer

4. Which two items are characteristics of the Gold Layer? (Choose two.)

a) Historical lineage
b) Raw Data
c) Normalised
d) De-normalised
e) Read-optimized

5. A data engineer team has decided to implement a new data platform on Databricks and is currently deciding how to store each kind of data on each data layer.
What is the appropriate layer and data pairing for medallion architecture?

a) Silver Layer - Raw data from deposit account application
b) Bronze Layer - Summary of cash deposit amount for each country and city
c) Silver Layer - Cleansed master customer data
d) Gold Layer - Deduplicated money transfer transaction

6. A data engineer has been provided a PySpark DataFrame named df with columns product and revenue. The data engineer needs to compute complex aggregations to determine each product's total revenue, average revenue, and transaction count.
Which code snippet should the data engineer use?

a)
```
from pyspark.sql import functions as F
aggregated_df = df.groupBy("product")
.agg(F.sum("revenue")
.alias("total_revenue"),
F.avg("revenue").alias("avg_revenue"),
F.count("*").alias("transaction_count"))
```

b)
```
aggregated_df = df.groupBy("product").agg(
"sum(revenue)",
"avg(revenue)",
"count(revenue)" )
```

c)
```
from pyspark.sql import functions as F
aggregated_df = df.select("product", "revenue")
.groupBy("product")
.agg(F.sum("revenue"),
F.mean("revenue"))
```

d)
```
aggregated_df = df.groupBy("product")
.agg({"revenue": "sum",
"revenue": "avg",
"revenue": "count"})
```

7. A data engineer is developing an ETL process based on Spark SQL. The execution fails. The data engineer checks the Spark UI and can see the ERRORS as follows:
"java.lang.OutofMemoryError: Java heap space"
Which two corrective actions should the data engineer perform to resolve this issue? (Choose two.)

a) Narrow the filters in order to collect less data in the query
b) Upsize the worker nodes and activate autoshuffle partitions
c) Upsize the driver node and deactivate autoshuffle partitions
d) Cache the dataset in order to boost the query performance
e) Fix the shuffle partitions to 50 to ensure the allocation

8. A data engineer is working in a Python notebook on Databricks to process data, but notices that the output is not as expected. The data engineer wants to investigate the issue by stepping through the code and checking the values of certain variables during execution.
Which tool should the data engineer use to inspect the code execution and variables in real-time?

a) Cluster Logs
b) Job Execution Dashboard
c) Python Notebook Interactive Debugger
d) SQL Analytics

9. A data engineer is writing a script that is meant to ingest new data from cloud storage. In the event of the Schema change, the ingestion should fail. It should fail until the changes downstream source can be found and verified as intended changes.
Which command will meet the requirements?

a) failonNewColumns
b) none
c) rescue
d) addNewColumns

10. A data engineer needs to ingest from both streaming and batch sources for a firm that relies on highly accurate data. Occasionally, some of the data picked up by the sensors that provide a streaming input are outside the expected parameters. If this occurs, the data must be dropped, but the stream should not fail.
Which feature of Delta Live Tables meets this requirement?

a) Change Data Capture
b) Error Handling
c) Monitoring
d) Expectations

11. A data engineer has inherited a Databricks pipeline from a previous team. The pipeline is missing SLAs and costs more than the allotted budget. On analysis, it is noted that the cluster is not being fully utilized, and the dataset is getting skewed.
How should the data engineer resolve this issue?

a) Use coalesce() on the dataset to merge partitions and reduce skew.
b) Increase the number of executors for the job.
c) Repartition the dataset to have it be more optimally spread across all nodes.
d) Increase the executor memory for the job.

12. A data engineer streams customer orders into a Kafka topic (orders_topic) and is currently writing the ingestion script of a DLT pipeline. The data engineer needs to ingest the data from Kafka brokers to DLT using Databricks.
What is the correct code for ingesting the data?

a)
```
import dlt
@dlt.table(name="orders_raw",)
def orders_raw():
return (spark.readStream
.format("kafka")
.option("kafka.bootstrap.servers", "broker:9092")
.option("subscribe", "orders_topic")
.option("startingOffsets", "earliest")
.load())
```

b)
```
import dlt
@dlt.table(name = "orders_raw",)
def orders_raw():
return (spark.readStream
.format("kafka")
.option("kafka.bootstrap.servers", "broker:9092")
.option("subscribe", "orders_topic")
.option("startingOffsets", "earliest")
.load())
```

c)
```
CREATE LIVE TABLE orders_raw
AS SELECT
CAST(value AS STRING) AS json_data
FROM STREAM kafka.`broker:9092/orders_topic`;
```

d)
```
CREATE STREAMING LIVE TABLE orders_raw
AS SELECT
value:order_id AS order_id,
value:customer_id AS customer_id,
value:amount AS amount,
value:order_status AS order_status,
value:order_timestamp AS order_timestamp
FROM cloud_files("kafka://broker:9092/orders_topic", "json");
```

13. A data engineer has developed an ETL that produce a Delta managed table with liquid clustering feature activated as output. Several consumers are having issues regarding time delay when reading this table.
How could the Data Engineer be sure about the OPTIMIZE command has been executed explicitly?

a) Check the system table system.storage.predictive_optimization operations_history

b) Use SHOW TABLES EXTENDED to check the partitions columns used

c) Use DESCRIBE DETAIL table to see the file size and number of files for the table

d) Use DESCRIBE HISTORY table to check if exists any OPTIMIZE operation

14. A data engineer is working on a Databricks project that utilizes cloud storage. The data engineer wants to load several JSON files from containers on a storage account as soon as the file arrives within the storage account.
Which syntax should the data engineer follow to first load the files into a dataframe and check that it is working as expected using Python?

a)
```
df = spark.read.json ("input/path")
```

b)
```
df = spark.readStream
.format("json".load("input/path")
```

c)
```
df = spark.readstream
.format("cloud")
.option ("json").load("/input/path")
```

d)
```
df = spark.readStream
.format("cloudFiles")
.option ("cloudFiles.format",
"json")
.load("/input/path")
```

15. A data engineer is processing ingested streaming tables and needs to filter out NULL values in the order_datetime column from the raw streaming table orders_raw and store the results in a new table orders_valid using DLT.
Which code snippet should the data engineer use?

a)
```
CREATE OR REFRESH STREAMING TABLE orders_valid(
CONSTRAINT valid_date
EXPECT (order_datetime IS NOT NULL)
ON VIOLATION DROP ROW)
AS SELECT * FROM STREAM orders_raw;
```

b)
```
CREATE OR REFRESH STREAMING TABLE orders_valid(
CONSTRAINT valid_date EXPECT (order_datetime IS NOT NULL)
ON VIOLATION DROP ROW )
AS SELECT * FROM orders_raw;
```

c)
```
CREATE OR REFRESH STREAMING TABLE orders_valid
AS
SELECT *
FROM STREAM(orders_raw)
WHERE order_datetime IS NOT NULL;
```

d)
```
CREATE OR REPLACE STREAMING TABLE orders_valid (
FILTER (order_datetime IS NOT NULL) )
AS SELECT * FROM STREAM(orders_raw);
```

16. What is the functionality of AutoLoader in Databricks?

a) Auto Loader automatically ingests and processes new files from cloud storage, handling both batch and streaming data with support for schema evolution.

b) Auto Loader automatically ingests and processes new files from cloud storage, handling batch and streaming data with no support for schema evolution.

c) Auto Loader automatically ingests and processes new files from cloud storage, handling only streaming data with no support for schema evolution.

d) Auto Loader automatically ingests and processes new files from cloud storage, handling batch data with support for schema evolution.

17. A data engineer is using the Databricks OPTIMIZE command on a Delta table.
What happens when OPTIMIZE is run twice on the same table with the same data?

a) It has no effect because it is idempotent.

b) It changes the number of tuples per file significantly.

c) It further reduces file sizes by re-clustering the data.

d) It triggers a full liquid clustering process.

18. A data engineer is inspecting an ETL pipeline based on a Pyspark job that consistently encounters performance bottlenecks. Based on developer feedback, the data engineer assumes the job is low on compute resources. To pinpoint the issue, the data engineer observes the Spark UI and finds out the job has a high CPU time vs Task time.
Which course of action should the data engineer take?

a) High CPU time vs Task time means an under-utilized cluster. The data engineer may need to repartition data to spread the jobs more evenly throughout the cluster.

b) High CPU time vs Task time means efficient use of cluster and no change needed

c) High CPU time vs Task time means a CPU over-utilized job. The data engineer may need to consider executor and core tuning or resizing the cluster

d) High CPU time vs Task time means over-utilized memory and the need to increase parallelism

19. A data engineer needs to parse only png files in a directory that contains files with different suffixes.
Which code should the data engineer use to achieve this task?

a)
```
df = spark.readStream.format("cloudFiles") \
.option ("cloudFiles.format", "binaryFile") \
.append("/*.png")
```

b)
```
df = spark.readstream. format("cloudFiles") \
.option("cloudFiles.format", "binaryFile") \
.option ("pathGlobfilter", "*.png") \
.load(<base-path>)
```

c)
```
df = spark.readStream.format("cloudFiles") \
.option ("cloudFiles.format", "binaryFile") \
.option ("pathGlobfilter", "*.png") \ .append()
```

d)
```
df = spark.readstream.format("cloudFiles") \
.option ("cloudFiles.format", "binaryFile") \
.load("/*.png")
```

20. A data engineer has inherited a Databricks pipeline from a previous team. The pipeline is missing its SLAs, and an initial investigation has identified memory spills in Spark. These increased runtimes are also driving up costs. The data engineer needs to reduce the runtime without significantly increasing costs.
What should the data engineer do first to address this issue?

a) Tweak the "spark.sql.shuffle.partitions" configuration.
b) Enable autoscaling in the cluster to match the requirements
c) Use a photon enabled execution engine.
d) Ensure the cluster uses memory optimized node types.

21. A data engineer is facing performance bottlenecks in an e-commerce transactions Delta table. The table is a managed Unity Catalog table, and it uses partitioning and Z-ordering in its data layout scheme. The predictive optimization for Unity Catalog tables is also enabled. The table has a frequently changing query filter, and the data engineer does not observe a benefit of the Data Layout or the Predictive Optimization.
How should the data engineer fix the data layout bottlenecks?

a) Re-write the Data Layout with Liquid Clustering and cluster by the Z-Ordered columns.
b) Enable Delta Caching so that query results can be read through caches.
c) Tweak the Z-Order columns and run OPTIMIZE manually.
d) Switch the Data layout from Partition+Z-Ordering to Automatic Liquid Clustering.

22. A data engineer needs to optimize the data layout and query performance for an e-commerce transactions Delta table. The table is partitioned by "purchase_date" a date column which helps with time-based queries but does not optimize searches on user statistics "customer_id", a high-cardinality column.
The table is usually queried with filters on "customer_id" within specific date ranges, but since this data is spread across multiple files in each partition, it results in full partition scans and increased runtime and costs.
How should the data engineer optimize the Data Layout for efficient reads?

a) Alter table implementing liquid clustering on "customer_id" while keeping the existing partitioning.
b) Alter the table implementing liquid clustering by "customer_id" and "purchase_date".
c) Alter the table to partition by "customer_id".
d) Enable delta caching on the cluster so that frequent reads are cached for performance.

23. A Delta Live Table pipeline includes two datasets defined using STREAMING LIVE TABLE. Three datasets are defined against Delta Lake table sources using LIVE TABLE. The table is configured to run in Production mode using the Continuous Pipeline Mode.
What is the expected outcome after clicking Start to update the pipeline assuming previously unprocessed data exists and all definitions are valid?

a) All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will persist to allow for additional testing.
b) All datasets will be updated once and the pipeline will shut down. The compute resources will persist to allow for additional testing.
c) All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will be deployed for the update and terminated when the pipeline is stopped.
d) All datasets will be updated once and the pipeline will shut down. The compute resources will be terminated.

24. A data engineer has a Job with multiple tasks that runs nightly. Each of the tasks runs slowly because the clusters take a long time to start.
Which action can the data engineer perform to improve the start up time for the clusters used for the Job?

a) They can use endpoints available in Databricks SQL
b) They can use jobs clusters instead of all-purpose clusters
c) They can configure the clusters to autoscale for larger data sizes
d) They can use clusters that are from a cluster pool

25. A data engineer has a single-task Job that runs each morning before they begin working. After identifying an upstream data issue, they need to set up another task to run a new notebook prior to the original task.
Which approach can the data engineer use to set up the new task?

a) They can clone the existing task in the existing Job and update it to run the new notebook.
b) They can create a new task in the existing Job and then add it as a dependency of the original task.
c) They can create a new task in the existing Job and then add the original task as a dependency of the new task.
d) They can create a new job from scratch and add both tasks to run concurrently.

26. A single Job runs two notebooks as two separate tasks. A data engineer has noticed that one of the notebooks is running slowly in the Job's current run. The data engineer asks a tech lead for help in identifying why this might be the case.
Which approach can the tech lead use to identify why the notebook is running slowly as part of the Job?

a) They can navigate to the Runs tab in the Jobs UI to immediately review the processing notebook.
b) They can navigate to the Tasks tab in the Jobs UI and click on the active run to review the processing notebook.
c) They can navigate to the Runs tab in the Jobs UI and click on the active run to review the processing notebook.
d) They can navigate to the Tasks tab in the Jobs UI to immediately review the processing notebook.

27. A data engineer and data analyst are working together on a data pipeline. The data engineer is working on the raw, bronze, and silver layers of the pipeline using Python, and the data analyst is working on the gold layer of the pipeline using SQL. The raw source of the pipeline is a streaming input. They now want to migrate their pipeline to use Delta Live Tables.
Which of the following changes will need to be made to the pipeline when migrating to Delta Live Tables?

a) The pipeline will need to be written entirely in Python
b) The pipeline will need to stop using the medallion-based multi-hop architecture
c) The pipeline will need to be written entirely in SQL
d) The pipeline will need to use a batch source in place of a streaming source

28. A data engineer is working in a Databricks notebook to design and manage a batch ETL pipeline. The engineer is writing SQL and Python code to clean data, transform it, and join large datasets from different sources. The engineer wants to organize these steps into a structured process that can be run regularly and scheduled as part of a data pipeline.
Which Databricks notebook feature is applicable in the use case?

a) Real-time streaming support
b) Collaborative editing
c) Task workflows and job scheduling
d) Notebook version control

29. A data engineer needs to develop integration tests for an ETL process and deploy a version-controlled, packaged workflow into production using an external job scheduler.
Which tool should the data engineer use for this job?

a) Databricks Connect
b) Databricks Asset Bundles
c) Databricks Command Line Interface
d) Databricks Software Development Kit

30. Which Databricks asset bundle format is valid?

a)
```
resources:
  jobs:
    hello-job:
      name: hello-job
      tasks:
        - task_key: hello-task
          existing_cluster_id: 1234-567890-abcde123
          notebook_task:
            notebook_path: ./hello.py
```

b)
```
{
  "resources": {
    "jobs": {
      "name": "hello-job",
      "tasks": {
        "task_key: "hello-task",
        "existing_cluster_id": "1234-567890-abcde123",
        "notebook_task": {
          "notebook_path": ".hello.py"
        }
      }
    }
  }
}
```

c)
```
configuration = {
  "resources": {
    "jobs": {
      "name": "hello-job",
      "tasks": {
        "task_key: "hello-task",
        "existing_cluster_id": "1234-567890-abcde123",
        "notebook_task": {
          "notebook_path": ".hello.py"
        }
      }
    }
  }
}
```

d)
```
resources {
  jobs {
    name = "hello-job"
    tasks {
      task_key = "hello-task"
      existing_cluster_id = "1234-567890-abcde123"
      notebook_task {
        notebook_path = ".hello.py"
      }
    }
  }
}
```

31. A data engineer is designing an ETL pipeline to process both streaming and batch data from multiple sources. The pipeline must ensure data quality, handle schema evolution, and provide easy maintenance. The team is considering using Delta Live Tables (DLT) in Databricks to achieve these goals. They want to understand the key features and benefits of DLT that make it suitable for this use case.
Why is Delta Live Tables (DLT) an appropriate choice?

a) Automatic data quality checks, built-in support for schema evolution, and declarative pipeline development
b) Manual schema enforcement, high operational overhead, and limited scalability
c) Requires custom code for data quality checks, no support for streaming data, and complex pipeline maintenance
d) Supports only batch processing, no data versioning, and high infrastructure costs

32. A Data Engineer is building a simple data pipeline using Delta Live Tables (DLT) in Databricks to ingest customer data. The raw customer data is stored in a cloud storage location in JSON format. The task is to create a DLT pipeline that reads the raw JSON data and writes it into a Delta table for further processing.
Which code snippet will correctly ingest the raw JSON data and create a Delta table using DLT?

a)
```
import dlt
@dlt.table
def raw_customers():
return spark.read.format("csv").load("s3://my-bucket/raw-customers/")
```

b)
```
import dlt
@dlt.view
def raw_customers():
return spark.format.json("s3://my-bucket/raw-customers/")
```

c)
```
import dlt
@dlt.table
def raw_customers():
return spark.read.json("s3://my-bucket/raw-customers/")
```

d)
```
import dlt
@dlt.table
def raw_customers():
return spark.read.format("parquet").load("s3://my-bucket/raw-customers/")
```

33. A Python file is ready to go into production and the client wants to use the cheapest but most efficient type of cluster possible. The workload is quite small, only processing 10GBS of data with only simple joins and no complex aggregations or wide transformations.
Which cluster meets the requirement?

a) Interactive cluster
b) Job cluster with spot instances enabled
c) Job cluster with spot instances disabled
d) Job cluster with Photon enabled

34. A Databricks workflow fails at the last stage due to an error in a notebook. This workflow runs daily. The data engineer fixes the mistake and wants to rerun the pipeline. This workflow is very costly and time-intensive to run.
Which action should the data engineer do in order to minimise downtime and cost?

a) Re-run the entire workflow
b) Repair run
c) Restart the cluster
d) Switch to another cluster

35. A Databricks single-task workflow fails at the last task due to an error in a notebook. The data engineer fixes the mistake in the notebook.
What should the data engineer do to rerun the workflow?

a) Repair the task
b) Rerun the pipeline
c) Restart the cluster
d) Switch the cluster

36. What is the structure of an Asset Bundle?

a) A Docker image containing runtime environments and the source code of the assets
b) A compressed archive (ZIP) that solely contains workspace assets without any accompanying metadata
c) A single plain text file enumerating the names of assets to be migrated to a new workspace
d) A YAML configuration file that specifies the artifacts, resources, and configurations for the project

37. A data engineer is maintaining an ETL pipeline code with a GitHub repository linked to their Databricks account. The data engineer wants to deploy the ETL pipeline to production as a databricks workflow.
Which approach should the data engineer use?

a) Databricks Asset Bundles (DAB) + GitHub Integration
b) Maintain workflow_config.json and deploy it using Databricks CLI
c) Maintain workflow_config.json and deploy it using Terraform
d) Manually create and manage the workflow in UI

38. A data engineer works for an organization that must meet a stringent Service Level Agreement (SLA) that demands minimal runtime errors and high availability for its data processing pipelines. The data engineer wants to avoid the operational overhead of managing and tuning clusters.
Which architectural solution will meet the requirements?

a) Use an auto-scaling cluster configured and monitored by the user.
b) Implement a hybrid approach with scheduled batch jobs on custom cloud VMs.
c) Deploy a dedicated, manually managed cluster optimized by in-house IT staff.
d) Utilize Databricks serverless compute that automatically optimizes resources and abstracts cluster management.

39. A data engineer wants to reduce costs and optimize cloud spending. The data engineer has decided to use Databricks Serverless for lowering cloud costs while maintaining existing SLAs.
What is the first step in migrating to Databricks Serverless?

a) Legacy Ingestion pipelines that include ingestion from sources API's, files, JDBC/ODBC connections
b) A frequently running and efficient Python-based data transformation pipeline compatible with the latest Databricks runtime and Unity Catalog
c) A frequently running and efficient Scala-based data transformation pipeline compatible with the latest Databricks runtime and Unity Catalog
d) Low frequency BI Dashboarding and Adhoc SQL Analytics

40. A data engineering project involves processing large batches of data on a daily schedule using ETL. The jobs are resource-intensive and vary in size, requiring a scalable, cost-efficient compute solution that can automatically scale based on the workload.
Which compute approach will satisfy the needs described?

a) Job Cluster
b) Dedicated Cluster
c) All-Purpose Cluster
d) Databricks SQL Serverless

41. A data engineer is maintaining a data pipeline. Upon data ingestion, the data engineer notices that the source data is starting to have a lower level of quality. The data engineer would like to automate the process of monitoring the quality level.
Which of the following tools can the data engineer use to solve this problem?

a) Auto Loader
b) Unity Catalog
c) Delta Lake
d) Delta Live Tables

42. A company uses Delta Sharing to collaborate with partners across different cloud providers and geographic regions.
What will result in additional costs due to cross-region or egress fees?

a) Sharing data within the same cloud provider and region
b) Transferring data via Delta Sharing across clouds and across different geographic regions
c) Accessing Delta Sharing data using a VPN within the same data center
d) Utilizing Delta Sharing for internal data analytics within a single cloud environment

43. An organization has data stored across multiple external systems, including MySQL, Amazon Redshift, and Google BigQuery. The data engineer wants to perform analytics without ingesting directly into Databricks, ensuring unified governance and minimizing data duplication.
Which feature of Databricks enables querying these external data sources while maintaining centralized governance?

a) Delta Lake
b) Lakehouse Federation
c) MLflow
d) Databricks Connect

44. An organization needs to share a dataset stored in its Databricks Unity Catalog with an external partner who uses a different data platform that is not Databricks. The goal is to maintain data security and ensure the partner can access the data efficiently.
Which method should the data engineer use to securely share the dataset with the external partner?

a) Using Delta Sharing with the open sharing protocol
b) Exporting data as CSV files and emailing them
c) Using a third-party API to access the Delta table
d) Databricks-to-Databricks Sharing

45. A data engineer is reviewing the documentation on audit logs in Databricks for compliance purposes and needs to understand the format in which audit logs output events. How are events formatted in Databricks audit logs?

a) In Databricks, audit logs output events in a JSON format.
b) In Databricks, audit logs output events in a CSV format.
c) In Databricks, audit logs output events in an XML format.
d) In Databricks, audit logs output events in a plain text format.

46. A data engineer is managing a data pipeline in Databricks, where multiple Delta tables are used for various transformations. The team wants to track how data flows through the pipeline, including identifying dependencies between Delta tables, notebooks, jobs, and dashboards. The data engineer is utilizing the Unity Catalog lineage feature to monitor this process.
How does Unity Catalog's data lineage feature support the visualization of relationships between Delta tables, notebooks, jobs, and dashboards?

a) Unity Catalog lineage visualizes dependencies between Delta tables, notebooks, and jobs, but does not provide column-level tracing or relationships with dashboards.

b) Unity Catalog lineage only supports visualizing relationships at the table level and does not extend to notebooks, jobs, or dashboards.

c) Unity Catalog lineage provides an interactive graph that tracks dependencies between tables and notebooks but excludes any job-related dependencies or dashboard visualizations.

d) Unity Catalog provides an interactive graph that visualizes the dependencies between Delta tables, notebooks, jobs, and dashboards, while also supporting column-level tracking of data transformations.

47. A data engineer is configuring Unity Catalog in Databricks and needs to assign a role to a user who should have the ability to grant and revoke privileges on various data objects within a specific schema, but should not have read/write access over the schema or its objects.
Which role should the data engineer assign to this user?

a) Table Owner

b) Catalog Owner

c) Schema Owner

d) USE catalog/schema privilege on the schema

48. A company is collaborating with a partner that does not use Databricks but needs access to a large historical dataset stored in Delta format. The data engineer needs to ensure that the partner can access the data securely, without the need for them to set up an account, and with read-only access.
How should the data be shared?

a) Share the dataset by exporting it to a CSV file and manually transferring the file to the partner's system.

b) Grant your partner access to your Databricks workspace and assign them full write permissions to the Delta table, enabling them to modify the dataset.

c) Share the dataset using Unity Catalog, ensuring that both teams have full write access to the data within the same organization.

d) Share the dataset using Delta Sharing, which allows your partner to access the data using a secure, read-only URL without requiring a Databricks account, ensuring that they cannot modify the data.

49. A data engineer at a company that uses Databricks with Unity Catalog needs to share a collection of tables with an external partner who also uses a Databricks workspace enabled for Unity Catalog. The data engineer decides to use Delta Sharing to accomplish this.
What is the first piece of information the data engineer should request from the external partner to set up Delta Sharing?

a) The IP address of their Databricks workspace

b) The name of their Databricks cluster

c) The sharing identifier of their Unity Catalog metastore

d) Their Databricks account password

50. A data engineer needs to provide access to a group named manufacturing-team. The team needs privileges to create tables in the quality schema.
Which set of SQL commands will grant a group named manufacturing-team to create tables in a schema named production with the parent catalog named manufacturing with the least privileges?

a)
```
GRANT CREATE TABLE ON SCHEMA manufacturing.quality To manufacturing-team;
GRANT USE SCHEMA ON SCHEMA manufacturing.quality To manufacturing- team;
GRANT USE CATALOG ON CATALOG manufacturing TO manufacturing-team;
```

b)
```
GRANT USE TABLE ON SCHEMA manufacturing.quality TO manufacturing-team;
GRANT USE SCHEMA ON SCHEMA manufacturing.quality TO manufacturing-team;
GRANT USE CATALOG ON CATALOG manufacturing TO manufacturing-team;
```

c)
```
GRANT CREATE TABLE ON SCHEMA manufacturing.quality TO manufacturing-team;
GRANT CREATE SCHEMA ON SCHEMA manufacturing.quality TO manufacturing-
team;
GRANT CREATE CATALOG ON CATALOG manufacturing TO manufacturing-team;
```

d)
```
GRANT CREATE TABLE ON SCHEMA manufacturing.quality TO manufacturing-team;
GRANT CREATE SCHEMA ON SCHEMA manufacturing.quality TO manufacturing-
team;
GRANT USE CATALOG ON CATALOG manufacturing TO manufacturing-team;
```

51. A data engineer is configuring Delta Sharing for a multi-team project where teams from different departments will need to access shared data. The data engineer has successfully created a Unity Catalog metastore and is now setting up the Delta Share. The goal is to ensure that internal teams can access the shared data with full permissions, while external partners can only read the data. Which action should the Data Engineer take to configure the sharing correctly?

a) Create a Delta Share, set up a secure access URL for internal teams and external partners, and distribute the URL to provide them access to the shared data.

b) Create a Delta Share, add the internal team's tables and views, and assign READ/WRITE permissions to both external partners and internal teams.

c) Assign READ permissions to external partners through the Delta Share and READ/WRITE permissions to internal teams, and ensure the correct tables and views are shared.

d) Grant READ permissions to external partners and READ/WRITE permissions to internal teams through the Delta Share.

52. A data engineer manages multiple external tables linked to various data sources. The data engineer wants to manage these external tables efficiently and ensure that only the necessary permissions are granted to users for accessing specific external tables.
How should the data engineer manage access to these external tables?

a) Set up Azure Blob Storage permissions at the container level, allowing access to all external tables.

b) Create a single user role with full access to all external tables and assign it to all users.

c) Grant permissions on the Databricks workspace level, which will automatically apply to all external tables.

d) Use Unity Catalog to manage access controls and permissions for each external table individually.

53. An organization plans to share a large dataset stored in a Databricks workspace on AWS with a partner organization whose Databricks workspace is hosted on Azure. The data engineer wants to minimize data transfer costs while ensuring secure and efficient data sharing.
Which strategy will reduce data egress costs associated with cross-cloud data sharing?

a) Migrating the dataset to Cloudflare R2 object storage before sharing

b) Configure VPN connection between AWS and Azure for faster data sharing

c) Using Delta Sharing without any additional configurations

d) Sharing data via pre-signed URLS without monitoring egress costs

54. A data engineer is setting up access control in Unity Catalog and needs to ensure that a group of data analysts can query tables but not modify data. Which permission should the data engineer grant to the data analysts?

a) ALL PRIVILEGES
b) MODIFY
c) SELECT
d) INSERT

55. What Databricks feature can be used to check the data sources and tables used in a workspace?

a) Use the lineage feature to visualize a graph that highlights where the table is used only in reports.
b) Use the lineage feature to visualize a graph that highlights where the table is used only in notebooks.
c) Use the lineage feature to visualize a graph that shows all dependencies, including where the table is used in notebooks, other tables, and reports.
d) Do not use the lineage feature as it only tracks activity from the last 3 months and will not provide full details on dependencies.

56. A data engineer is getting a partner organization up to speed with Databricks account. Both the teams share some business use cases. The data engineer has to share some of your Unity-Catalog managed delta tables and the notebook jobs creating those tables with the partner organization.
How can the data engineer seamlessly share the required information?

a) Zip all the code and share via email and allow data ingestion from your data lake
b) Share required datasets and notebooks via Delta Sharing. Manage permissions via Unity Catalog.
c) Data and Notebooks can be shared simply using Unity Catalog.
d) Share access to codebase via Github and allow them to ingest datasets from your Datalake.

57. A data engineer needs to audit who made changes to a Delta table in the last 7 days. Which SQL statement returns the record of past operations, including timestamps, users, and operation types?

a) DESCRIBE DETAIL my_table
b) DESCRIBE HISTORY my_table
c) SHOW TBLPROPERTIES my_table
d) SELECT * FROM 'delta.my_table/_delta_log/'

58. A data engineer is attempting to grant a user access to a view. They successfully run GRANT SELECT ON VIEW sales catalog.market_data.summary TO analyst_group. However, the analyst still receives an "Insufficient Permissions" error when trying to query the view. In which order should the engineer verify the Hierarchy of Securable Objects to ensure the analyst has the necessary usage permissions?

a) CATALOG -> SCHEMA -> VIEW
b) METASTORE -> TABLE -> VIEW
c) STORAGE CREDENTIAL -> EXTERNAL LOCATION -> VIEW
d) WORKSPACE -> CLUSTER -> VIEW

59. A data engineer needs to support a Unity Catalog Delta table catalog. schema. sales that is currently registered as an external table. New requirements state:
+ Automatic file cleanup after a drop is handled by the platform. » Minimal downtime during migration while keeping the same table name, permissions, and history.
+ Enforced name-based access through Unity Catalog (no path-based access).
Which action should the data engineer take to meet the requirements?

a) Run ALTER TABLE catalog.schema.sales SET MANAGED to convert and meet the goals
b) Run DROP TABLE catalog.schema.sales PURGE to delete files and meet goals
c) Run ALTER TABLE catalog.schema.sales SET LOCATION to switch external paths now
d) Create a new managed table with CTAS and migrate workloads to that table today

60. A data engineering team ingests customer transaction data from three enterprise sources (SQL database, S3, and Kafka stream) into Unity Catalog tables. The data must maintain lineage and support compliance audits requiring historical snapshots.

Which Lakeflow Connect configuration satisfies both the governance and audit requirements?

a) Single Lakeflow connector with all three sources, direct Delta writes, and disabled table versioning to reduce storage costs.

b) Separate connectors per source with CDC, individual Unity Catalog schemas per source, and table versioning enabled for audit compliance.

c) Separate connectors with full-load sync only, archival of historical data, and external tables for audit queries.

d) Single connector with batch scheduling, shared Unity Catalog schema, and incremental snapshots stored externally.

61. A governance team is evaluating whether to use Unity Catalog attribute-based access control (ABAC) policies or manually applied row filters and column masks to protect sensitive data across their catalog. Why should the team use ABAC policies instead of manually applied row filters and column masks?

a) ABAC policies use governed tags to dynamically match tables and columns, so a single policy defined on a catalog automatically applies to all current and future tables within that catalog without per-table configuration.

b) ABAC policies allow row filters and column masks to be written in Python and Scala directly, whereas manually applied filters and masks can only use SQL user-defined functions.

c) ABAC policies can be applied to views and materialized views in addition to tables, whereas manually applied row filters and column masks can only be applied to tables.

d) ABAC policies support applying multiple distinct row filters to the same table for a single user at query time, whereas manually applied row filters are limited to one filter per table.

62. A data engineer needs to enforce row-level security on main. secure.events (region STRING, event_id STRING). Members of the account group all_regions need to see all rows. All other users need to see only rows where region = 'EU'.
Which SQL sequence satisfies the requirement?

a)
```
CREATE FUNCTION main.secure.region_filter(region STRING)
RETURN IF(is_account_group_member('all_regions'), true, region = 'EU');
```

b)
```
ALTER TABLE main.secure.events
SET ROW FILTER main.secure.region_filter ON (region);
CREATE FUNCTION main.secure.region_filter(region STRING)
RETURN IF(is_account_group_member('all_regions'), true, region = 'EU');
```

c)
```
ALTER VIEW main.secure.events
SET ROW FILTER main.secure.region_filter ON (region);
CREATE FUNCTION main.secure.region_filter(region STRING)
RETURN IF(is_account_group_member('all_regions'), true, region = 'EU');
```

d)
```
ALTER TABLE main.secure.events
ALTER COLUMN region
SET MASK main.secure.region_filter;
CREATE FUNCTION main.secure.region_filter(region STRING)
RETURN IF(is_account_group_member('all_regions'), true, region = 'EU');

ALTER TABLE main.secure.events
SET ROW FILTER main.secure.region_filter USING COLUMNS (region);
```

63. A data engineer uses the Databricks workspace UI with Unity Catalog enabled. In Data Explorer, they select catalog corp marketing, then schema campaigns, and see table email_stats. The engineer must let the growth-analysts group read email_stats from their SQL warehouses, but not create, alter, or delete any objects in corp_marketing or campaigns.
Which action sequence meets the requirement?

a) 1. On schema campaigns, open Permissions and grant USE SCHEMA to growth analysts 2. On the table email stats, open Permissions and grant SELECT and MODIFY to growth-analysts

b) 1. On catalog corp marketing, open Permissions and grant USE CATALOG to growth-analysts 2. On schema campaigns, open Permissions and grant USE SCHEMA to growth-analysts 3. On the table email stats, open Permissions and grant SELECT to growth-analysts

c) 1. On catalog corp marketing, open Permissions and grant USE CATALOG and CREATE SCHEMA to growth-analysts 2. On schema campaigns, open Permissions and grant USE SCHEMA to growth-analysts 3. On the table email _stats, open Permissions and grant SELECT to growth-analysts

d) 1. On schema campaigns, open Permissions and grant USE SCHEMA and SELECT to growth-analysts 2. On the table email stats, open Permissions and grant SELECT to growth-analysts

64. A data engineer needs to configure an external location and a storage credential so that a new bronze ingestion pipeline can land data in cloud storage governed by Unity Catalog. The engineer has CREATE TABLE permission on schemas but receives a permission error when attempting to create the external location.
Which role is required to unblock this setup, the principle of least privilege?

a) Grant CREATE EXTERNAL LOCATION on both the metastore and the referenced storage credential so the engineer can define the external location.

b) Grant ALL PRIVILEGES on the catalog so the engineer can create external locations under its schemas.

c) Promote the engineer to Workspace Admin so they can manage objects and jobs.

d) Assign the engineer as Metastore Admin because only Metastore Admins can create external locations.

65. A data engineer is onboarding a new bronze ingestion pipeline in Databricks with Unity Catalog. The team wants Databricks to handle storage layout, apply platform optimizations over time, and simplify lifecycle management so that when a table is dropped, its underlying data is also cleaned up according to Databricks-managed retention policies.
Which table type should the data engineer create for these ingestion tables?

a) Managed tables so that Unity Catalog manages both metadata and underlying data lifecycle.

b) External tables with a LOCATION pointing to an external volume for full control of file layout.

c) Foreign tables federated from an external catalog to delegate optimization to the source system

d) Temporary views over files to avoid table-level governance and lifecycle coupling.

66. A data engineer is decommissioning a sandbox schema in Unity Catalog. Some tables are ephemeral staging outputs that can be safety removed entirely, but a few tables point at shared cloud storage used by downstream jobs outside Databricks. The engineer must avoid deleting any shared files when cleaning up catalog objects.
How does Unity Catalog behave when dropping Managed vs External tables?

a) DROP all tables; Databricks will only remove metadata for both managed and external tables.

b) DROP managed tables that are ephemeral and DROP external tables; files for both remain for 7 days.

c) DROP managed staging tables to remove data and metadata, and DROP external tables to remove only metadata

d) DROP external tables first to delete their files, then DROP managed tables to keep their files for recovery.

67. A data engineer wants to delegate day-to-day permission management for the schema main marketing to the mkt-admins group, without making them workspace admins. They should be able to grant and revoke privileges for other users on objects within that schema.
Which approach aligns with UC's ownership and privilege model?

a) Grant MANAGE permissions on the metastore to mkt-admins, which allows managing privileges for all schemas and tables globally.

b) Make mkt-admins a workspace-level admins group, then SELECT on main.marketing to allow privilege delegation.

c) Transfer ownership of the schema main.marketing to mkt-admins; owners can manage privileges on the schema and its contained objects.

d) Grant USE SCHEMA on main.marketing and MODIFY on all tables to mkt-admins, which enables managing grants within the schema.

68. Which Databricks feature can be used to check the data sources and tables used in a workspace?

a) Use the lineage feature to visualize a graph that shows all dependencies, including where the table is used in notebooks, other tables, and reports.

b) Use the lineage feature to visualize a graph that highlights where the table is used only in reports.

c) Use the audit logs feature to review user activities, such as who queried or modified a table, instead of visualizing dependencies between data objects.

d) Use the lineage feature to visualize a graph that highlights where the table is used only in notebooks.

69.

| | version | timestamp | operation |
|---|---|---|---|
| 1 | 8 | 2024-04-22T14:33:31.000 | OPTIMIZE |
| 2 | 7 | 2024-04-22T14:33:16.000 | MERGE |
| 3 | 6 | 2024-04-22T14:33:06.000 | DELETE |
| 4 | 5 | 2024-04-22T14:32:58.000 | UPDATE |
| 5 | 4 | 2024-04-22T14:32:47.000 | WRITE |
| 6 | 3 | 2024-04-22T14:32:44.000 | WRITE |
| 7 | 2 | 2024-04-22T14:32:23.000 | WRITE |
| 8 | 1 | 2024-04-22T14:32:20.000 | WRITE |
| 9 | 0 | 2024-04-22T14:31:49.000 | CREATE TABLE |

The Delta transaction log for the 'students' tables is shown using the 'DESCRIBE HISTORY students' command. A Data Engineer needs to query the table as it existed before the UPDATE operation listed in the log. Which command should the Data Engineer use to achieve this? (Choose two.)

a)
```
SELECT * FROM students@v4
```

b)
```
SELECT * FROM students TIMESTAMP AS OF '2024-04-22T14:32:47.000+00:00
```

c)
```
SELECT * FROM students FROM HISTORY VERSION AS OF 3
```

d)
```
SELECT * FROM students VERSION AS OF 5
```

e)
```
SELECT * FROM students TIMESTAMP AS OF '2024-04-22T14:32:58.000+00:00'
```

70. A data engineer needs to access the view created by the sales team, using a shared cluster. The data engineer has been provided usage permissions on the catalog and schema. In order to access the view created by sales team. What are the minimum permissions the data engineer would require in addition?

a) Needs SELECT permission on the VIEW and the underlying TABLE.

b) Needs SELECT permission only on the VIEW

c) Needs ALL PRIVILEGES on the VIEW

d) Needs ALL PRIVILEGES at the SCHEMA level

71. Which two conditions are applicable for governance in Databricks Unity Catalog? (Choose two.)

a) You can have more than 1 metastore within a databricks account console but only 1 per region.

b) Both catalog and schema must have a managed location in Unity Catalog provided metastore is not associated with a location

c) You can have multiple catalogs within metastore and 1 catalog can be associated with multiple metastore

d) If catalog is not associated with location, it's mandatory to associate schema with managed locations

e) .If metastore is not associated with location, it's mandatory to associate catalog with managed locations

72. dentify the impact of ON VIOLATION DROP ROW and ON VIOLATION FAIL UPDATE for a constraint violation. A data engineer has created an ETL pipeline using Delta Live table to manage their company travel reimbursement detail, they want to ensure that the if the location details has not been provided by the employee, the pipeline needs to be terminated. How can the scenario be implemented?

a) CONSTRAINT valid_location EXPECT (location = NULL)

b) CONSTRAINT valid_location EXPECT (location != NULL) ON VIOLATION FAIL UPDATE

c) CONSTRAINT valid_location EXPECT (location != NULL) ON DROP ROW

d) CONSTRAINT valid_location EXPECT (location != NULL) ON VIOLATION FAIL

73. In a healthcare provider organization using Delta Lake to store electronic health records (EHRs), a data analyst needs to analyze a snapshot of the patient_records table from two weeks ago before some recent data corrections were applied. What approach should the Data Engineer take to allow the analyst to query that specific prior version?

a) Truncate the table to remove all data, then reload the data from two weeks ago into the truncated table for the analyst to query.

b) Identify the version number corresponding to two weeks ago from the Delta transaction log, share that version number with the analyst to query using VERSION AS OF syntax, or export that version to a new Delta table for the analyst to query.

c) Restore the table to the version from two weeks ago using the RESTORE command, and have the analyst query the restored table.

d) Use the VACUUM command to remove all versions of the table older than two weeks, then the analyst can query the remaining version

74. A dataset has been defined using Delta Live Tables and includes an expectations clause:
CONSTRAINT valid timestamp EXPECT (timestamp > '2020-01 01') ON VIOLATION DROP ROW
What is the expected behavior when a batch of data containing data that violates these constraints is processed?

a) Records that violate the expectation cause the job to fail.

b) Records that violate the expectation are added to the target dataset and flagged as invalid in a field added to the target dataset.

c) Records that violate the expectation are dropped from the target dataset and recorded as invalid in the event log.

d) Records that violate the expectation are added to the target dataset and recorded as invalid in the event log.

75. A data engineer has three tables in a Delta Live Tables (DLT) pipeline. They have configured the pipeline to drop invalid records at each table. They notice that some data is being dropped due to quality concerns at some point in the DLT pipeline. They would like to determine at which table in their pipeline the data is being dropped. Which approach can the data engineer take to identify the table that is dropping the records?

a) They can set up separate expectations for each table when developing their DLT pipeline.
b) They can navigate to the DLT pipeline page, click on the “Error” button, and review the present errors.
c) They can set up DLT to notify them via email when records are dropped.
d) They can navigate to the DLT pipeline page, click on each table, and view the data quality statistics.

76. A new data engineering team team has been assigned to an ELT project. The new data engineering team will need full privileges on the database customers to fully manage the project.
Which of the following commands can be used to grant full permissions on the database to the new data engineering team?

a) GRANT USAGE ON DATABASE customers TO team;
b) GRANT ALL PRIVILEGES ON DATABASE team TO customers;
c) GRANT SELECT PRIVILEGES ON DATABASE customers TO teams;
d) GRANT SELECT CREATE MODIFY USAGE PRIVILEGES ON DATABASE customers TO team:
e) GRANT ALL PRIVILEGES ON DATABASE customers TO team:

77. A dataset has been defined using Delta Live Tables and includes an expectations clause: CONSTRAINT valid_timestamp EXPECT (timestamp > ‘2020-01 01’) ON VIOLATION DROP ROW.
What is the expected behavior when a batch of data containing data that violates these constraints is processed?

a) Records that violate the expectation are dropped from the target dataset and loaded into a quarantine table.
b) Records that violate the expectation cause the job to fail.
c) Records that violate the expectation are dropped from the target dataset and recorded as invalid in the event log.
d) Records that violate the expectation are added to the target dataset and recorded as invalid in the event log.

78. Which single Databricks CLI command deploys local bundle assets to the target workspace specified in the configuration file?

a) databricks bundle init && databricks bundle deployment
b) databricks bundle plan && databricks bundle run
c) databricks bundle deploy
d) databricks bundle validate

79. A Databricks Job with four sequential tasks is executed. The job failed at Task 3. After the root cause is fixed, the data engineer needs to complete the workflow without re-running the successful Tasks 1 and 2.
Which action in the Job Run details page allows the engineer to resume execution from the point of failure?

a) Repair run
b) Reset job
c) Clone task
d) Run now

80. A data engineering team runs multiple daily batch jobs. The team needs to run the jobs on a fixed overnight schedule with consistent performance and controlled resource usage. The team wants to minimize operational overhead and reduce idle cluster costs.
Which compute option should the team use?

a) SQL warehouses
b) Classic all-purpose compute
c) Serverless jobs compute
d) Classic jobs compute

81. A data engineer is building a Silver layer pipeline that should only execute after the Bronze layer table has been successfully updated with new records.
Which trigger type in Databricks Jobs is specifically designed to orchestrate this dependency by reacting to changes in the upstream Unity Catalog table?

a) Table update trigger
b) File arrival trigger
c) Scheduled trigger
d) Continuous trigger

82. A data engineer has a bundle with a job resource keyed as et1_job in databricks.yml. After running 'databricks bundle deploy', the job appears in the workspace but has not executed.
Which command triggers execution of the deployed job?

a) databricks jobs submit --json
b) databricks bundle run etl_job
c) databricks bundle deploy --run
d) databricks bundle execute etl_job

83. A data engineer triggers a scheduled job but finds that the new run was not executed. The run history shows the run was skipped with a concurrency related queue message.
Which configuration should the engineer investigate?

a) The SQL warehouse permissions for the task
b) The task-level timeout_seconds configuration
c) The job-level max_concurrent_runs setting and whether queueing is enabled
d) The cluster spot-instance availability in the cloud provider

84. A data engineer is preparing a Declarative Automation Bundle (formerly Databricks Asset Bundle) to deploy a Lakeflow pipeline.
To ensure the pipeline is deployed to the correct environment, where should the workspace specific configurations, such as the host URL and root storage path, be defined within the bundle project?

a) In a global environment variable on the local development machine
b) Within the Python source code of the Lakeflow pipeline
c) In the targets section of the databricks.ym1 file
d) Inside the README . md file of the bundle root directory

85. A data engineer needs Task C to run ONLY if Task A succeeds AND Task B fails.
Which dependency configuration implements this conditional logic?

a) Task C depends on Task A (All done) and Task B (All done) with a condition widget.
b) Task C depends on Task A (Success) with Task B listed as a negative dependency.
c) Task C uses two dependency chains: Task A with All succeeded, Task B with at least one failed.
d) Create a Python task that checks dbutils.jobs.taskValues from Tasks A and B before running Task C.

86. A data engineering team runs nightly batch ETL pipelines that transform large volumes of data stored in Delta Lake. The workloads are predictable and run on a fixed schedule. The team wants automatic scaling and cost efficiency, paying only for compute used during job execution.
Which Databricks compute option is appropriate for this workload?

a) An all-purpose compute cluster configured with autoscaling.
b) A Databricks SQL warehouse configured for serverless execution.
c) A single-node compute cluster configured without autoscaling.
d) A job compute cluster triggered by a scheduled workflow.

87. A notebook task check_volume calculates a daily data volume and writes it to a Lakeflow Jobs taskValue named record_count. If the count is greater than 1 million, the job should run high *volume*pipeline; otherwise, it should run normal_pipeline.
Which control-flow feature should be used?

a) Add an I£/else condition task that reads the record_count task value and routes to different downstream tasks.
b) Use a For each task over the record_count and stop when it is below the threshold.
c) Configure additional retries for check_volume until the count exceeds the threshold.
d) Configure Run if conditions based purely on task status.

88. A data engineer is designing a Databricks Job that includes a Notebook task for data cleansing and a SQL Query task for generating a summary report. The summary report must only execute after the cleansing task completes successfully.
How should the dependency be configured in the Lakeflow Jobs task graph?

a) Select the cleansing task as a 'Run if' dependency in the summary report task configuration.
b) Set both tasks to run on the same schedule with a five-minute start delay for the report.
c) Configure the cleansing task to use an 'On Failure' trigger to start the summary report.
d) Place both tasks in the same notebook and use a standard Python function call.

89. A data engineer has finished developing a new Declarative Automation Bundle (formerly Databricks Asset Bundle) for a streaming pipeline. They need to ensure the configuration is syntactically correct, push the code to the workspace, and finally trigger the pipeline to verify it runs as expected.
Which sequence of Databricks CLI commands should the engineer use to perform these steps in the correct order?

a) databricks bundle init -> databricks bundle deploy -> databricks bundle run
b) databricks bundle validate -> databricks bundle deploy -> databricks bundle run
c) databricks bundle validate -> databricks bundle sync -> databricks bundle execute
d) databricks bundle check -> databricks bundle upload -> databricks bundle start

90. A company has a strict 15-minute Service Level Agreement (SLA) for updating its currency exchange dashboard. The source data arrives in small increments every few minutes. The data engineering team needs a triggering strategy for their Lakeflow Job that keeps end-to-end latency within the SLA while minimizing compute cost and DBU consumption.
Which triggering strategy for the Lakeflow Job is the recommended way to meet this 15-minute SLA?

a) Use a Scheduled trigger set to run every 12 minutes, with the streaming task configured to use Trigger .AvailableNow.
b) Use a Scheduled trigger set to run every 1 minute to ensure the SLA is never at risk.
c) Use a File Arrival trigger configured to fire every time a single JSON file lands in the bucket.
d) Use a Continuous trigger to ensure data is processed as soon as it arrives.

**Answer Keys**

1. a)
2. b)
3. c) Bronze
4. d) De-normalised , e) Read-optimized
5. c) Silver Layer - Cleansed master customer data
6. b)
7. a) Narrow the filters in order to collect less data in the query , b) Upsize the worker nodes and activate autoshuffle partitions
8. c) Python Notebook Interactive Debugger
9. a) failonNewColumns
10. d) Expectations
11. c) Repartition the dataset to have it be more optimally spread across all nodes.
12. c)
13. d) Use DESCRIBE HISTORY table to check if exists any OPTIMIZE operation
14. d)
15. a)
16. a) Auto Loader automatically ingests and processes new files from cloud storage, handling both batch and streaming data with support for schema evolution.
17. a) It has no effect because it is idempotent.
18. c) High CPU time vs Task time means a CPU over-utilized job. The data engineer may need to consider executor and core tuning or resizing the cluster
19. b)
20. a) Tweak the "spark.sql.shuffle.partitions" configuration.
21. d) Switch the Data layout from Partition+Z-Ordering to Automatic Liquid Clustering.
22. b) Alter the table implementing liquid clustering by "customer_id" and "purchase_date".
23. c) All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will be deployed for the update and terminated when the pipeline is stopped.
24. d) They can use clusters that are from a cluster pool
25. b) They can create a new task in the existing Job and then add it as a dependency of the original task.
26. c) They can navigate to the Runs tab in the Jobs Ul and click on the active run to review the processing notebook.
27. a) The pipeline will need to be written entirely in Python
28. c) Task workflows and job scheduling
29. b) Databricks Asset Bundles
30. a)
31. a) Automatic data quality checks, built-in support for schema evolution, pipeline development
32. c)
33. b) Job cluster with spot instances enabled

and declarative

34. b) Repair run

35. a) Repair the task

36. d) A YAML project configuration file that specifies the artifacts, resources, and configurations for the

37. a) Databricks Asset Bundles (DAB) + GitHub Integration

38. d) Utilize Databricks serverless compute that automatically optimizes resources and abstracts cluster management.

39. b) A frequently running and efficient Python-based data transformation pipeline compatible with the latest Databricks runtime and Unity Catalog

40. a) Job Cluster

41. d) Delta Live Tables

42. b) Transferring data via Delta Sharing across clouds and across different geographic regions

43. b) Lakehouse Federation

44. a) Using Delta Sharing with the open sharing protocol

45. a) In Databricks, audit logs output events in a JSON format.

46. b) Unity Catalog lineage only supports visualizing relationships at the table level and does not extend to notebooks, jobs, or dashboards.

47. c) Schema Owner

48. d) Share the dataset using Delta Sharing, which allows your partner to access the data using a secure, read-only URL without requiring a Databricks account, ensuring that they cannot modify the data.

49. c) The sharing identifier of their Unity Catalog metastore

50. a)

51. c) Assign READ permissions to external partners through the Delta Share and READ/WRITE permissions to internal teams, and ensure the correct tables and views are shared.

52. d) Use Unity Catalog to manage access controls and permissions for each external table individually.

53. c) Using Delta Sharing without any additional configurations

54. c) SELECT

55. c) Use the lineage feature to visualize a graph that shows all dependencies, including where the table is used in notebooks, other tables, and reports.

56. b) Share required datasets and notebooks via Delta Sharing. Manage permissions via Unity Catalog.

57. b) DESCRIBE HISTORY my_table

58. a) CATALOG -> SCHEMA -> VIEW

59. a) Run ALTER TABLE catalog.schema.sales SET MANAGED to convert and meet the goals

60. b) Separate connectors per source with CDC, individual Unity Catalog schemas per source, and table versioning enabled for audit compliance.

61. a) ABAC policies use governed tags to dynamically match tables and columns, so a single policy defined on a catalog automatically applies to all current and future tables within that catalog without per-table configuration.

62. a)

63. b) 1. On catalog corp marketing, open Permissions and grant USE CATALOG to growth-analysts 2. On schema campaigns, open Permissions and grant USE SCHEMA to growth-analysts 3. On the table email stats, open Permissions and grant SELECT to growth-analysts

64. a) Grant CREATE EXTERNAL LOCATION on both the metastore and the referenced storage credential so the engineer can define the external location.

65. a) Managed tables so that Unity Catalog manages both metadata and underlying data lifecycle.

66. c) DROP managed staging tables to remove data and metadata, and DROP external tables to remove only metadata

67. c) Transfer ownership of the schema main.marketing to mkt-admins; owners can manage privileges on the schema and its contained objects.

68. a) Use the lineage feature to visualize a graph that shows all dependencies, including where the table is used in notebooks, other tables, and reports.

69. a) , b)

70. b) Needs SELECT permission only on the VIEW

71. a) You can have more than 1 metastore within a databricks account console but only 1 per region. , e) .If metastore is not associated with location, it's mandatory to associate catalog with managed locations

72. b) CONSTRAINT valid_location EXPECT (location != NULL) ON VIOLATION FAIL UPDATE

73. b) Identify the version number corresponding to two weeks ago from the Delta transaction log, share that version number with the analyst to query using VERSION AS OF syntax, or export that version to a new Delta

74. c) Records that violate the expectation are dropped from the target dataset and recorded as invalid in the event log.

75. d) They can navigate to the DLT pipeline page, click on each table, and view the data quality statistics.

table for the analyst to query.

76. e) GRANT ALL PRIVILEGES ON DATABASE customers TO team:

77. c) Records that violate the expectation are dropped from the target dataset and recorded as invalid in the event log.

78. c) databricks bundle deploy

79. a) Repair run

80. c) Serverless jobs compute

81. a) Table update trigger

82. b) databricks bundle run etl_job

83. c) The job-level max_concurrent_runs setting and whether queueing is enabled

84. c) In the targets section of the databricks.ym1 file

85. c) Task C uses two dependency chains: Task A with All succeeded, Task B with at least one failed.

86. d) A job compute cluster triggered by a scheduled workflow.

87. a) Add an l£/else condition task that reads the record_count task value and routes to different downstream tasks.

88. a) Select the cleansing task as a 'Run if' dependency in the summary report task configuration.

89. b) databricks bundle validate -> databricks bundle deploy -> databricks bundle run

90. a) Use a Scheduled trigger set to run every 12 minutes, with the streaming task configured to use Trigger .AvailableNow.